



Childhood Pulmonary Inflammatory Myofibroblastic Tumors Treatment (PDQ®)– Health Professional Version

Incidence and Histology

Inflammatory myofibroblastic tumors (IMTs) occur throughout the body, but the lungs are the most commonly involved organs. IMTs are one of the most frequent lung tumors in children, accounting for between 16% and 38% of cases in various series.[1-3]

The biology of IMT is variable, with the potential for local recurrences and rare distant metastases.[4] The *ALK* locus (located on chromosome 2p23) is rearranged in approximately 50% of IMT cases. *ALK* rearrangements can involve various different genes.[5,6] Other potentially targetable fusions have been reported in a smaller fraction of IMTs, including *ROS1*, *NTRK3*, *RET*, and *PDGFRB* fusions.[6,7]

References

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Clinical Presentation and Diagnostic Evaluation

Inflammatory myofibroblastic tumors (IMTs) present with a large, peripherally located lobulated mass with a lower lobe predominance. Chest wall, vascular, and mediastinal invasion may be seen. Calcification is occasionally present. Enhancement is heterogeneous on contrast-enhanced computed tomography. Magnetic resonance imaging findings are variable, but IMTs may be isointense relative to skeletal muscle.

Treatment of Childhood Pulmonary Inflammatory Myofibroblastic Tumors

Treatment options for pulmonary inflammatory myofibroblastic tumors (IMTs) include the following:[\[1-3\]](#)

1. [Surgery](#).
2. [Targeted therapy](#).

Surgery

If possible, surgical resection is the treatment of choice. Patients with completely resected tumors have an excellent prognosis.

Targeted Therapy

Patients with unresectable or recurrent tumors may respond to crizotinib if the *ALK* variant is present and crizotinib administration is followed by complete or incomplete resection. Treatment with ceritinib and entrectinib have also produced objective responses.

Evidence (targeted therapy):

1. **Crizotinib.**
 - a. One study included 14 patients with IMTs who were treated with crizotinib.[\[4\]](#)
[\[Level of evidence C3\]](#)

- Five patients had complete responses, seven patients had partial responses, and the remaining two patients had stable disease.
- No patients experienced a relapse when this article was published.

b. An extensive review confirmed that crizotinib was effective in children who had IMTs (with various tumor sites).[5]

The U.S. Food and Drug Administration (FDA) approved crizotinib for use in patients aged 1 year and older with unresectable, recurrent, or refractory *ALK*-positive IMTs.

2. **Ceritinib.** In a multicenter phase I study, seven of ten patients with IMTs had objective responses to ceritinib.[6]
3. **Entrectinib.** In a phase I/II study of entrectinib, two patients with IMTs and *ALK* fusions experienced a complete response and a partial response, respectively.[7]
4. **Alectinib.** A case report described the successful treatment of a patient with an IMT and a *FN1::ALK* gene fusion using alectinib, a second-generation *ALK* inhibitor.[8]

A retrospective, international, multicenter study analyzed patients younger than 21 years with *ROS1*-altered IMTs who were enrolled in either the European paediatric Soft Tissue Sarcoma Study Group (EpSSG) NRSTS-2005 study or the Soft Tissue Sarcoma Registry. Primary surgery was recommended if a microscopic radical resection without disfigurement was feasible. Of the 19 patients, 12 received neoadjuvant systemic therapy as first-line treatment (high-dose steroids, *n* = 2; vinorelbine/vinblastine with methotrexate, *n* = 6; *ROS1* inhibitors, *n* = 8). With a median follow-up of 2.8 years, seven patients had an event. The 3-year event-free survival rate was 41% (95% CI, 11%–71%), and the overall survival rate was 100%. While many patients in this series received crizotinib, the specific *ROS1* inhibitor used for each patient was not specified.[9]

For more information about the treatment of this tumor, see the [Inflammatory myofibroblastic tumor and epithelioid inflammatory myofibroblastic sarcoma](#) section in Childhood Soft Tissue Sarcoma Treatment.

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Treatment Options Under Clinical Evaluation for Childhood Pulmonary Inflammatory Myofibroblastic Tumors

Information about National Cancer Institute (NCI)–supported clinical trials can be found on the [NCI website](#). For information about clinical trials sponsored by other organizations, see the [ClinicalTrials.gov website](#).

Special Considerations for the Treatment of Children With Cancer

Cancer in children and adolescents is rare, although the overall incidence has slowly increased since 1975.^[1] Children and adolescents with cancer should be referred to medical centers that have a multidisciplinary team of cancer specialists with experience treating the cancers that occur during childhood and adolescence. This multidisciplinary team approach incorporates

the skills of the following pediatric specialists and others to ensure that children receive treatment, supportive care, and rehabilitation to achieve optimal survival and quality of life:

- Primary care physicians.
- Pediatric surgeons.
- Pathologists.
- Pediatric radiation oncologists.
- Pediatric medical oncologists and hematologists.
- Ophthalmologists.
- Rehabilitation specialists.
- Pediatric oncology nurses.
- Social workers.
- Child-life professionals.
- Psychologists.
- Nutritionists.

For specific information about supportive care for children and adolescents with cancer, see the summaries on [Supportive and Palliative Care](#).

The American Academy of Pediatrics has outlined guidelines for pediatric cancer centers and their role in the treatment of children and adolescents with cancer.[2] At these centers, clinical trials are available for most types of cancer that occur in children and adolescents, and the opportunity to participate is offered to most patients and their families. Clinical trials for children and adolescents diagnosed with cancer are generally designed to compare potentially better therapy with current standard therapy. Other types of clinical trials test novel therapies when there is no standard therapy for a cancer diagnosis. Most of the progress in identifying curative therapies for childhood cancers has been achieved through clinical trials. Information about ongoing clinical trials is available from the [NCI website](#).

Dramatic improvements in survival have been achieved for children and adolescents with cancer. Between 1975 and 2020, childhood cancer mortality decreased by more than 50%.[3-5] Childhood and adolescent cancer survivors require close monitoring because side effects of cancer therapy may persist or develop months or years after treatment. For information about the incidence, type, and monitoring of late effects in childhood and adolescent cancer survivors, see [Late Effects of Treatment for Childhood Cancer](#).

Childhood cancer is a rare disease, with about 15,000 cases diagnosed annually in the United States in individuals younger than 20 years.[6] The U.S. [Rare Diseases Act of 2002](#) defines a

rare disease as one that affects populations smaller than 200,000 people in the United States. Therefore, all pediatric cancers are considered rare.

The designation of a rare tumor is not uniform among pediatric and adult groups. In adults, rare cancers are defined as those with an annual incidence of fewer than six cases per 100,000 people. They account for up to 24% of all cancers diagnosed in the European Union and about 20% of all cancers diagnosed in the United States.[7,8] In children and adolescents, the designation of a rare tumor is not uniform among international groups, as follows:

- A consensus effort between the European Union Joint Action on Rare Cancers and the European Cooperative Study Group for Rare Pediatric Cancers estimated that 11% of all cancers in patients younger than 20 years could be categorized as very rare. This consensus group defined very rare cancers as those with annual incidences of fewer than two cases per 1 million people. However, three additional histologies (thyroid carcinoma, melanoma, and testicular cancer) with incidences of more than two cases per 1 million people were also included in the very rare group due to a lack of knowledge and expertise in the management of these tumors.[9]
- The Children's Oncology Group defines rare pediatric cancers as those listed in the International Classification of Childhood Cancer subgroup XI, which includes thyroid cancers, melanomas and nonmelanoma skin cancers, and multiple types of carcinomas (e.g., adrenocortical carcinomas, nasopharyngeal carcinomas, and most adult-type carcinomas such as breast cancers and colorectal cancers).[10] These diagnoses account for about 5% of the cancers diagnosed in children aged 0 to 14 years and about 27% of the cancers diagnosed in adolescents aged 15 to 19 years.[4]

Most cancers in subgroup XI are either melanomas or thyroid cancers, with other cancer types accounting for only 2% of the cancers diagnosed in children aged 0 to 14 years and 9.3% of the cancers diagnosed in adolescents aged 15 to 19 years.

These rare cancers are extremely challenging to study because of the relatively few patients with any individual diagnosis, the predominance of rare cancers in the adolescent population, and the small number of clinical trials for adolescents with rare cancers.

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Latest Updates to This Summary (12/03/2024)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Treatment of Childhood Pulmonary Inflammatory Myofibroblastic Tumors (IMTs)

Added [alectinib](#) to the list of targeted therapies used for patients with IMTs. Also added text to state that a case report described the successful treatment of a patient with an IMT and a *FN1::ALK* gene fusion using alectinib, a second-generation ALK inhibitor (cited Fujiki et al. as reference 8).

Added [text](#) about the results of a retrospective, international, multicenter study that analyzed patients younger than 21 years with *ROS1*-altered IMTs who were enrolled in either the European paediatric Soft Tissue Sarcoma Study Group NRSTS-2005 study or the Soft Tissue Sarcoma Registry (cited Schoot et al. as reference 9).

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About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the treatment of childhood pulmonary inflammatory myofibroblastic tumors. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

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Changes to the summaries are made through a consensus process in which Board members evaluate the strength of the evidence in the published articles and determine how the article should be included in the summary.

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